PAPER CODE: BCE-C514
SEMESTER EXAMINATION- DECEMBER 2024
CLASS: B.TECH. SEMESTER: V
SUBJECT: CSE, FET
PAPER TITLE: CLOUD COMPUTING

Time: 3 hour **Max. Marks: 70**

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any ***five*** questions in about 150 words each. Each question carries six marks. **(5 X 6 = 30 Marks)**

		CO	BL
Q-1	What is a Cloud? Discuss the various characteristics of clouds.	1	1
Q-2	Briefly compare grid and cloud computing.	1	1
Q-3	Explain the different service models (IaaS, PaaS, SaaS) in cloud computing. How do they cater to varying business needs?	2	2
Q-4	Explain how encapsulation and isolation work in virtualization and their importance in cloud computing.	2	3
Q-5	Discuss the importance of API security in the context of cloud services.	3	2
Q-6	What features in PaaS platforms are critical for supporting modern DevOps practices?	3	2
Q-7	How does a common cloud management platform streamline resource allocation and monitoring?	3	3
Q-8	What are vertical clouds, and how do they address specific industry requirements?	3	2
Q-9	How does compliance with regulatory requirements influence cloud deployment decisions?	4	4
Q-10	What are the key layers of the cloud security reference model, and how do they address different security needs?	4	3

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any ***four*** questions in detail. Each question carries 10 marks. (4 x 10 = Marks)

		CO	BL
Q-1	Evaluate the role of public clouds in commercial environments. Discuss their advantages, challenges, and common use cases.	4	5
Q-2	Examine the importance of elasticity and scalability in cloud computing.	3	4
Q-3	Compare single-tenancy and multi-tenancy in cloud environments.	3	3
Q-4	Explain the concept of Database as a Service (DBaaS) and Monitoring as a Service (MaaS).	2	2
Q-5	Explain the difference between System Virtual Machines (VMs), Process Virtual Machines (VMs), and Virtual Machine Monitors (VMM).	3	2
Q-6	Compare and contrast hypervisors like Xen, KVM, VMware, VirtualBox, and Hyper-V.	3	2
Q-7	Explain the role of Elastic Compute Cloud (EC2) in AWS.	5	2
Q-8	Discuss Aneka's platform for building and managing distributed cloud applications. How does it contribute to cloud computing?	4	3